Methods of proof – 4

Invariants, monovariants and double counting

Homework

13 сентября 2026 года

Theoretical minimum. An invariant proves impossibility, a monovariant proves termination, and double counting proves identities and inequalities. To find an invariant, look at what the allowed operation does to the simplest functionals: sum, sum of squares, parity, residue, sign of a permutation, rank, determinant; the right invariant is usually the one that the operation multiplies by a constant or leaves untouched. To find a monovariant, look for a bounded integer quantity that strictly decreases, or for a continuous quantity with a sign-definite derivative. Double counting is the habit of computing one number in two ways; when the count is over incidences it produces an inequality, and when it is over a set of pairs it produces a bijective proof. A generating function is the same bookkeeping written algebraically.

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1. Numbers $1, 2, \dots, n$ are written on a board; one may erase two numbers and write their difference. Determine for which n the last remaining number can be zero, and find the invariant responsible.

2. Prove that a 10×10 board cannot be tiled by 1×4 pieces, by a colouring invariant, and determine all $m \times n$ boards tileable by $1 \times k$ pieces.

3. In a group of people every two are friends or enemies. Each step, a person with more enemies than friends changes sides: his friends become his enemies and vice versa. Prove that the process terminates, by exhibiting a monovariant, and estimate the number of steps.

4. Prove that $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$ by double counting, and $\sum_k k\binom{n}{k} = n2^{n-1}$ in two ways; then give a bijective proof of Vandermonde's identity.

5. Prove that in any finite set system the number of pairs (x, A) with $x \in A$ counted in two ways gives Fisher's inequality for a design, and deduce the bound on the number of blocks.

6. Prove that the trace of a nilpotent matrix is zero by using an invariant of similarity, and deduce that $AB - BA = E$ is impossible over a field of characteristic zero (compare with Linear algebra – 12).

7. Prove that the parity of the number of inversions is invariant under a transposition, and use it to show that the alternating group has index two; then apply the sign invariant to a puzzle of your choice.

8. A finite set of points in the plane is such that the midpoint of any two of them also belongs to the set. Prove that the set has one point, and give the monovariant proof and the extremal proof; compare with the lattice version (Convex – 10).

9. Prove that a sequence of moves in which a number is replaced by the sum of its digits terminates, find the invariant modulo 9, and prove the general statement about the digital root.

10. Prove Euler's formula $V - E + F = 2$ for planar graphs by an induction that is really a monovariant argument, and deduce that K_5 is not planar by double counting the incidences between faces and edges.